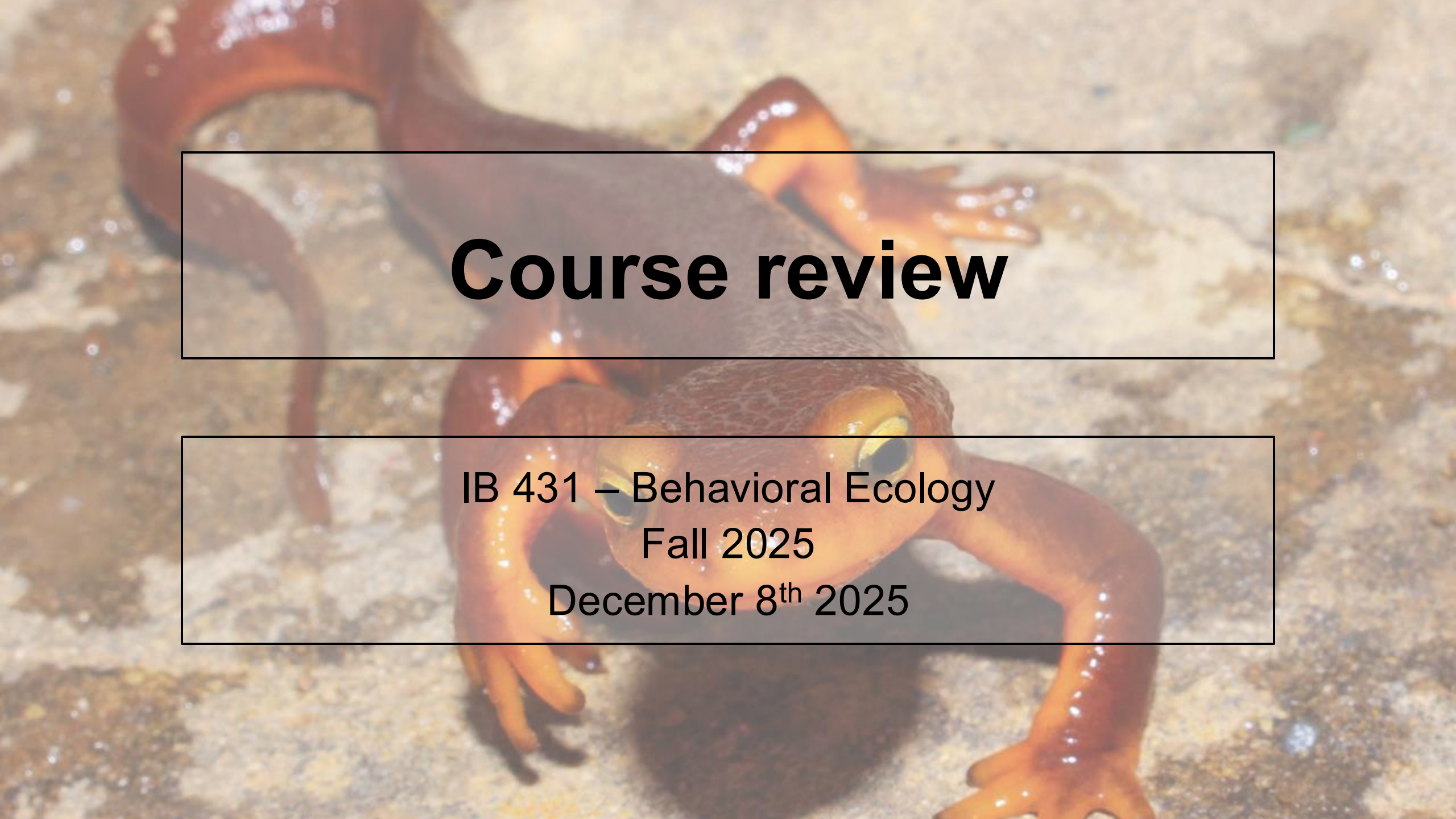




Course review

IB 431 – Behavioral Ecology
Fall 2025
December 8th 2025

Announcements

- Extended office hour times posted on Canvas
 - Can also email to set up another time
- Final proposal due Monday 12/15 at 11:59pm
 - Focus on feedback you got on second draft
- Final exam questions due Thursday 12/18 at 11:59pm
 - –10% for every day an assignment is late

Tips for the final exam

- Read clearly and address all parts of the question
- Questions are open-ended and have multiple answers – bringing up material from lectures, readings, or your own research will help you develop a strong answer
- Drawing connections between concepts/topics is great
- Don't be afraid to challenge the questions – if there is an assumption hidden within the question (and there are for some), you could make that part of your answer
- Work with peers and instructors – you need your own answer, but feedback is encouraged (this is a key part of science)
- Focus on quality of answer, not length

Outline

- I. Review of class
- II. Communicating behavioral ecology
- III. Discuss final exam questions with groups

Variation

- Genetic
- Phenotypic
- Environmental



Wall lizards (*Podarcis* sp.)

Interactions

- With abiotic environment
- Within individuals (i.e., phenotypic correlations)
- Within species
- Across species



Why do animals do what they do?

- Genes / gene expression
- Hormones
- Brain / neural mechanisms
- Learned behaviors
- Responding to other organisms
- Acquire resources
- Drive to reproduce
- Ancestors were exposed to conditions favoring a behavior
- Climatic / weather conditions



Studying behavioral evolution

- Identify variation in behavior
- Hypothesize connection with ecological function, environmental response, etc. (fitness) and design study to compare behavior across conditions
- **Measure heritability of behavior**



Game theory and equilibrium

- Equilibrium occurs when payoff of one strategy equal to payoff of alternative strategy
- Evolutionary stable strategy (ESS)
 - Strategy or strategies where no outside strategy (e.g., from a mutation) will be able to propagate in the population
 - For natterjack toads: ESS is equilibrium of callers/satellites



Recipient's fitness

Increases

Decreases

Increases

**Actor's
fitness**

Decreases

Cooperation

Selfishness

Altruism

Spite

Cooperation	Selfishness
Altruism	Spite

Using objective definitions of behaviors

- Establish clear definition of behavior(s) with ethogram
 - Avoid anthropomorphization
 - Ensure behavior is discrete
- Taking video of behavior
- Blinded observations
- Multiple individuals observe same behavior and see if they have similar results



Why do animals form groups?

- Foraging / hunting
- Avoiding predators
- Acquire information
- Find mates / care for offspring
- Family groups
- Thermoregulation
- Incidentally (i.e., high population density)



Kin selection

- Meerkats typically live in closely related groups
 - Example of Hamilton's rule

Cooperation is favored when: **$rB - C > 0$**

r = relatedness between altruist and recipient

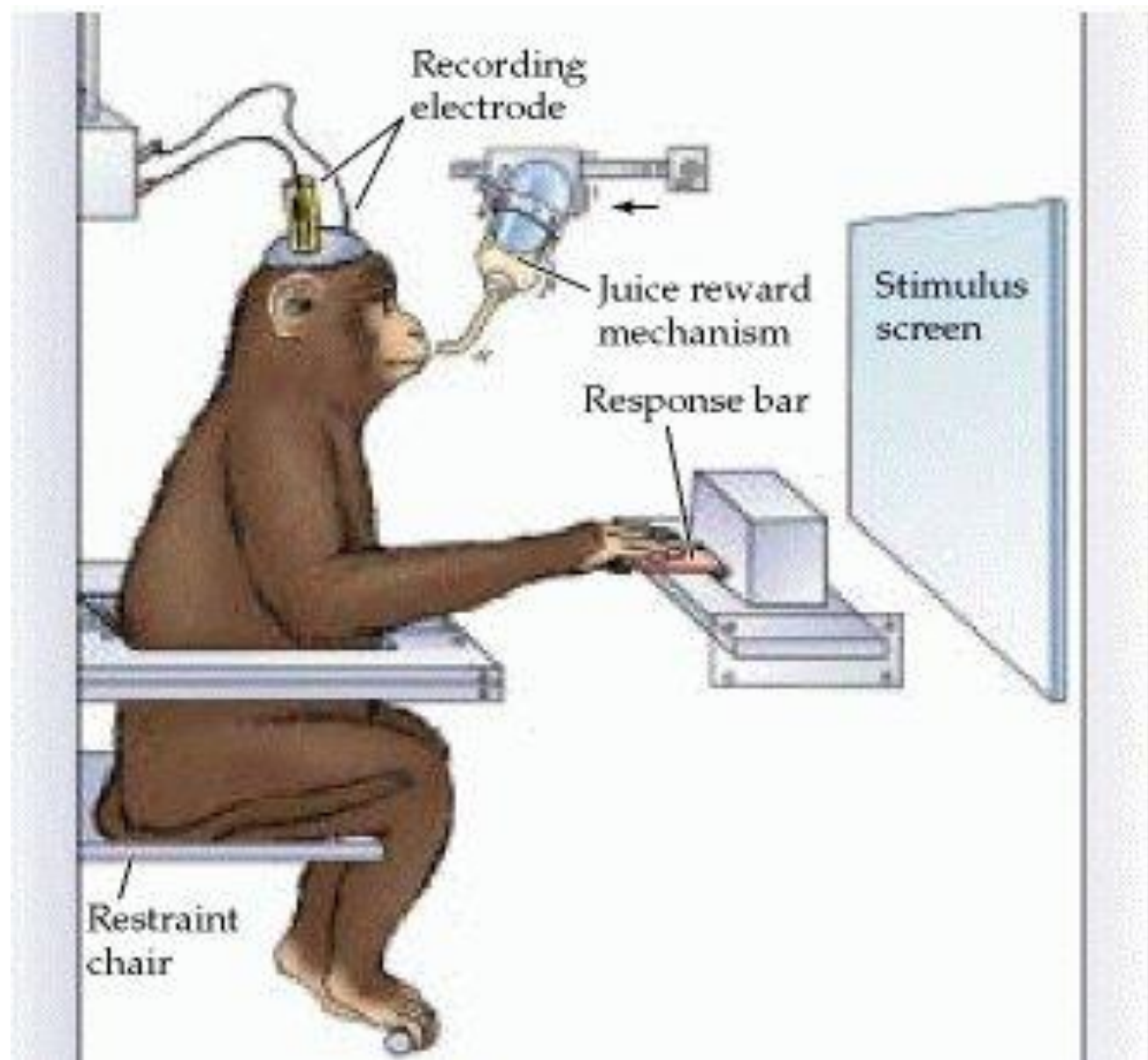
B = fitness benefit to recipient

C = fitness cost to the altruist

What about cooperation in unrelated individuals?

- Cooperation as byproduct of selfish behavior
- Demand for the same resources brings groups together
 - Reptiles basking in sunny areas
 - Zebras drinking from a swimming hole





Collective behavior

- **Movement**
 - Cohesion / coordination
- **Task performance**
 - Nest construction, foraging, parental care
- **Interactions**
 - Social networks
 - Division of labor / social roles
 - Collective decision making



Behavioral trade-offs

- Behavioral decisions where animals must choose to increase one aspect of fitness at the cost of another
 - Predation risk vs. foraging
 - Social information vs. increased aggression



Life history trade-offs and development

- Life history strategy often shapes risk-taking or exploration
 - Faster pace of life = might favor increased risk-taking, exploration
- Fairy wrens – more active, aggressive birds died sooner



Hall et al 2015

Behavioral development: Rat Park experiment

- Given enriched environment (other rats, toys, large cage) – rats prefer normal water over drug water



Communication

- Basically everything in behavioral ecology relies on communication
 - Game theory
 - Courtship, mating, care
 - Aggression /territoriality
 - Social interactions and spread of information
 - Predation



Approaches to behavioral conservation

- Managing conditions for breeding and courtship
- Facilitating dispersal (e.g., via wildlife bridges)
- Maintaining stable predator-prey population levels
- Establishing habitats that align with species preferences



Biases in behavioral ecology research

- Geographic biases
- Taxonomic biases
- Identity / background of authors
- Bias for positive / significant results
- Perspectives

Policy issues connect with animal behavior

- Conservation and climate change
- Ecosystem services
- Agriculture
- Endangered species
- Fisheries
- Animal welfare
- Public health



General takeaways

- Animal behavior is key to our understanding of the evolution of life on Earth
- It is deeply entrenched in ecological and ecosystem processes
- It has shaped human society and human history
- Animals are cool



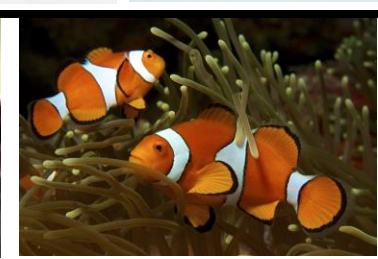
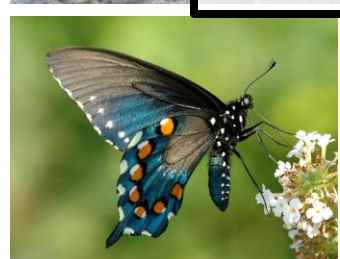
Communicating behavioral ecology

- Take a few minutes to discuss with your tables:
 - How do you communicate behavioral ecology? Why should the general public care about it?
 - What are some of the most interesting behaviors that you could share with someone to get them interested in the field?
 - Many of you are may be pursuing careers in other fields. How does your behavioral ecology knowledge apply to other careers/fields?



30 organisms

- 10 mammals
- 6 birds
- 4 arthropods
- 3 fish
- 2 plants
- 1 amphibian
- 1 jellyfish
- 1 mollusk
- 1 reptile
- 1 worm



Thank you all for your effort and engagement with this class!

I am happy to continue supporting you through letters of recommendation, feedback on personal statements or research proposals, or any questions about grad school, research, or career stuff.

In case I ever lose my Illinois email you can contact me at: **neumannkevin10@gmail.com**

